

Michael Dowsett

Machine learning researcher and software developer with hands-on experience designing and training CNN and R-CNN architectures for signal classification and parameter regression. Practical background in full-stack development with Python, PyTorch, and modern web frameworks.

✉ mikeydowsett@proton.me

☎ 913-251-1020

📍 KC, USA

🌐 mikeydowsett.com

PROJECTS

RF Signal Analysis – Multi-Head R-CNN

Spring 2026

- Designed and trained a multi-head R-CNN on 500,000 complex IQ waveforms (real/imaginary channels), with dual classification heads and a regression head predicting pulse width, time delay, and pulse repetition interval.
- Achieved 100% accuracy on signal type classification, 99.98% on pulse count classification, and an R^2 of 0.9841 on regression outputs across SNR-filtered test data.
- Applied standardized regression targets and tuned training with Cross Entropy and MSE loss functions across 20 epochs, batch size 128, learning rate 0.005.

Lofty – Fediverse Social Media Manager

Summer 2025

- Implemented OAuth login flows for Mastodon, Bluesky, Pixelfed, and Lemmy, enabling secure and reliable authentication.
- Engineered a FastAPI backend for handling user actions and Supabase integrations for tracking and analytics.
- Developed and deployed responsive Vue/Quasar front-end components with a focus on clean UI and smooth UX.

WORK EXPERIENCE

M.I.D.E

Student Research Assistant

04/2026 - Present

- Designed and trained CNN and R-CNN models with multiple classification heads and a regression head, selecting architectures best suited to the dataset and detection task.
- Fine-tuning pre-trained models for domain-specific tasks to improve output quality and relevance.

RoKC Enterprises

Front Desk/Key Holder

10/2025 - Present

- Managed member check-ins, processed retail and gear rentals, and provided a welcoming experience for climbers of all levels; handled opening/closing duties to ensure smooth gym operations.
- Assisted new climbers with orientations, answered questions about gym policies and climbing basics, and contributed to a positive, inclusive gym culture.

SKILLS

Python ●●●●●●

PyTorch ●●●●●●

Object-Oriented Programming ●●●●●●

- Languages: Python, C#, SQL, C++, TypeScript, Java, Bash
- Tools: Git, Linux, Google Colab, Machine Learning
- Libraries/Frameworks: PyTorch, NumPy, scikit-learn, FastAPI, Vue.js, .Net

EDUCATION

University of Missouri - Kansas City

Computer Science BS

12/2027

- Micro-credential: U.S. Department of Education Center of Excellence: AI-Empowered Spatial Computing

Johnson County Community College

Associates of Information Science - 3.7 GPA

05/2024

- Phi Theta Kappa